

Project Assignment: Elevator control and Human Machine Interface

Table of Contents

1 Main assignment requirements.....	1
1.1 Task Description.....	1
1.2 Project Overview.....	1
1.3 Design.....	2
1.4 Implementation.....	2
1.5 Documentation.....	2
1.6 Functionality: Control System (1-2 persons).....	2
1.7 Functionality: Human Machine Interface (HMI) (1-3 persons).....	3
1.8 System Architecture.....	3
2 Organisation.....	4
2.1 Control-HMI Communication.....	4
2.2 Software-Licenses.....	4
2.3 Infrastructure.....	4
3 Deliverables.....	4
4 Server Access.....	4
4.1 I/O List.....	5
4.2 Distances and sensor positions.....	6
4.3 Speeds.....	6
5 More information / links.....	6

1 Main assignment requirements

1.1 Task Description

Develop an elevator control system according to the description in this document. This requires the following steps:

1. Create a project overview with responsible persons
2. Design software components for control system and HMI (Human Machine Interface)
3. Implement and test the components
4. Document your work results

1.2 Project Overview

The project overview must contain the following information:

1. List of group participants (3-4)
2. Simple project plan with an estimation of hours per person with respect to a specific project domain (e.g. simulation, communication, database, control, component A/B/C).
3. Define one responsible person for each project domain (every group participant should be responsible for at least one project domain)

6 weeks of project duration amount to, e.g., 144 hours with 2 persons (6 ECTS ~ 12h/week).

1.3 Design

First, devise a software architecture by means of a method and notation of your choice. The notation can be in, e.g., UML, Text and/or Tables. A division into modules / components with well-defined interfaces facilitates build-up of the overall solution. Each component should provide an interface to interact with it from the HMI or, e.g., for testing purposes.

Interface variables should be writable in one direction, only. An interface variable/attribute may be defined by means of two variables, with one showing the current state, and the other being the writable interface to change this state. This is a suggestion and other ways to systematically structure interfaces are possible.

1.4 Implementation

Develop the Control- and HMI-Software according to your design and project overview.

1.5 Documentation

Document the finalized project. The documentation consists of:

1. Project overview
2. Design considerations
3. Optional test plan and test execution and result
4. Development effort by person and project domain according to your project overview. This must be based on a weekly, individual work report.
5. Evaluation of project evolution and potential improvements

1.6 Functionality: Control System (1-2 persons)

An elevator transports persons between several levels.

- 1) There are 4 levels.
- 2) The Motor can be controlled with 2 speeds in up- and down-direction, or it is turned off. Additional crawl speed can be used for fine adjustments.
- 3) Each level is associated with two adjacent cabin detection sensors in different distance, and for each approach direction.
- 4) One sensor detects the safe position to open the door.
- 5) The cabin should be driven at the highest speed. It may slow down, when approaching a level to be stopped at.
- 6) A person outside the elevator may choose to select her desired direction of travel. The elevator stops at her level, if it is traveling in the correct direction.
- 7) A person inside the elevator may choose her desired target level.
- 8) If the elevator is idle, it should begin to move into the direction of the first request (time).
- 9) The elevator continues in one direction, as long as there are requests in that direction, and as long as it did not reach the uppermost or lowermost level.
- 10) If at a stop on one level there are no more requests in the current direction of travel, but in the other direction, the direction of travel is changed.
- 11) Minimum stop time at a level is 6s + time for opening and closing the door.
- 12) In the cabin, there is an emergency stop button, which aborts/stops the motor at any position. After resetting the emergency state, the elevator continues to travel towards the next scheduled stop.

1.7 Functionality: Human Machine Interface (HMI) (1-3 persons)

All functions should be accessible via the HMI. The following requirements have to be fulfilled.

- 1) Operate all operation panels of the elevator (Level: Up/Down; Cabin-Panel; Supervisory Functions)
- 2) Logging of each stop at a particular level (Arrival, Departure, Door opening/open/closing/closed)
- 3) Logging of all alarms and modes of the elevator
- 4) All user interactions should be logged
- 5) There should be at least two levels of user access:
 - a. Supervisor: can operate all functions
 - b. Operator/User: may operate standard user panels, only
- 6) The current state of the elevator must be shown as a schematic view of the elevator. All 4 levels should be visible including door state. Operator panels should be visible and free for interaction.
- 7) Implement a function for a fairness statistic:
 - a. Simulate persons, who want to travel from one level to another.
 - b. Each person is situated in one of: start-level, elevator, target-level.
 - c. Each person is initially situated at a start-level.
 - d. On the start-level, a person requests the elevator in the direction of her target-level.
 - e. If the elevator stops at a persons start-level and if it is travelling in the right direction, the person moves into the elevator.
 - f. If the elevator stops at the target-level of a person in the elevator, the person moves into her target-level.
 - g. As soon as a person arrives at her target-level, the travel time from start- to target-level is calculated and evaluated for the statistic.
 - h. The statistics should contain at least: minimum, maximum, average travel time.
 - i. It must be possible, that at any point in time, one each level one person is waiting.
 - j. The capacity in terms of persons of the elevator is unlimited.

Section 7) is optional for a HMI group consisting of 2 persons.

1.8 System Architecture

The final system consists of

- 1) Elevator simulation on a PLC in the laboratory for automation technology. This is ready provided by the automation laboratory. Access:
 - a. Control via Modbus/TCP
 - b. State visualization via web-browser
- 2) Control program to be developed according to this task description and according to common understanding captured by Wiki
- 3) HMI with its configuration to be developed according to this task description and according to common understanding captured by Moodle/Wiki

- 4) Optional data base, communication system/service/server, depending on the common system architecture
- 5) The control program runs on a Raspberry Pi provided by and situated in the automation laboratory. The HMI runs on a different machine.

2 Organisation

The interface between HMI and control system should be as similar as possible, i.e., ideally, each HMI should be able to communicate with each control system – and vice versa.

Deadlines, documentation, etc. will be additionally published via Moodle.

A gitlab repository with wiki and issue tracker will be provided for each group.

Contact to the teacher:

see https://wiki.ei.htwg-konstanz.de/professoren_webseiten/krupp/start for

- Email
- Open office hours

2.1 Control-HMI Communication

The communication interface between the partner groups has to be specified as soon as possible in the gitlab wiki of the faculty of electrical engineering.

2.2 Software-Licenses

- CodeSys available free of charge for educational purpose
- Real-Time Java-VM (JamaicaVM)
- Standard-Java VM

2.3 Infrastructure

ea-pc111.ei.htwg-konstanz.de	Real-Time elevator simulation
ea-pc165.ei.htwg-konstanz.de	Standard-Linux Host with mosquitto, mysql Login: ssh, user: FHKN\{rzuser}

Raspberry pi installed in A231

AUT-Network

3 Deliverables

1. Project overview and description of the communication interface. The description must explain the basic protocol used, exchanged data type and message structure, and a state automaton and/or a sequence chart describing the order and type of messages exchanged. Submission in the gitlab wiki.
Deadline Tue 16.06.26
2. Plug-Fest / Hackathon: Tue 23.06.26 15:45-19:00h
3. Presentation of projects of group pairs C*-H*: Tue 30.06.26
4. Project source code and documentation submission on moodle
Deadline Fri 03.06.26

4 Server Access

Each control-system group may access their own elevator simulation. The web visualization of CodeSys enables your inspection of the system state by means of a web browser.

Simulation-GUI: <http://ea-pc111.ei.htwg-konstanz.de:8080/aufzug{group}simu.htm>

Control-GUI: <http://ea-pc111.ei.htwg-konstanz.de:8080/aufzug{group}ctrl.htm>

Replace {group} with your groups letter.

To control an elevator you have to implement a Modbus/TCP master and connect it to your assigned Modbus-Slave. A refresh rate of approx. 100ms is currently sufficient.

The Modbus Ports for each group are as follows:

Group D(C1): ea-pc111:505

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugdsimu.htm>

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugdctrl.htm>

Group E(C2): ea-pc111:506

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugesimu.htm>

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugectrl.htm>

Group F: ea-pc111:507

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugfsimu.htm>

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugfctrl.htm>

Group G: ea-pc111:508

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzuggsimu.htm>

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzuggctrl.htm>

Group H: ea-pc111:509

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzughsimu.htm>

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzughctrl.htm>

Group I: ea-pc111:510

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugisimu.htm>

<http://ea-pc111.ei.htwg-konstanz.de:8080/aufzugictrl.htm>

Access to the server is possible from the laboratory and from HTWG VPN.

4.1 I/O List

The I/O of the elevator is mapped to a Modbus I/O Image. (Codesys-Syntax)

```
P0reset    AT %QX0.0 : BOOL; // Simulation reset
P0dv2      AT %QX1.0 : BOOL; // motor setpoint down v2
P0dv1      AT %QX1.1 : BOOL; // motor setpoint down v1
P0uv1      AT %QX1.2 : BOOL; // motor setpoint up v1
P0uv2      AT %QX1.3 : BOOL; // motor setpoint up v2
P0dclose   AT %QX1.4 : BOOL; // motor close door
P0dopen    AT %QX1.5 : BOOL; // motor open door
P0v_crawlSelect AT %QW1 : INT (-5..5); // select crawl speed; -5..5 cm/s
```

```
PIs_l1sl   AT %IX0.1 : BOOL; // Level 1 safety lower
PIs_l1r     AT %IX0.2 : BOOL; // Level 1 reached
PIs_l1su    AT %IX0.3 : BOOL; // Level 1 safety upper
PIs_l1au    AT %IX0.4 : BOOL; // Level 1 approach upper
PIs_l2al    AT %IX1.0 : BOOL; // Level 2 approach lower
PIs_l2sl    AT %IX1.1 : BOOL; // etc.
PIs_l2r     AT %IX1.2 : BOOL;
PIs_l2su    AT %IX1.3 : BOOL;
PIs_l2au    AT %IX1.4 : BOOL;
PIs_l3al    AT %IX2.0 : BOOL;
PIs_l3sl    AT %IX2.1 : BOOL;
PIs_l3r     AT %IX2.2 : BOOL;
PIs_l3su    AT %IX2.3 : BOOL;
```

```

PIs_l3au AT %IX2.4 : BOOL;
PIs_l4al AT %IX3.0 : BOOL;
PIs_l4sl AT %IX3.1 : BOOL;
PIs_l4r AT %IX3.2 : BOOL;
PIs_l4su AT %IX3.3 : BOOL;
PIcycles AT %ID1 : UDINT; // PLC cycles since start
PIaufzugID AT %IW4 : UINT; // instance identifier (1..4)
PIs_dopened AT %IX10.0 : BOOL; // door is open
PIs_dclosed AT %IX10.1 : BOOL; // door is closed
PIm_ready AT %IX10.2 : BOOL; // motor ready
PIm_on AT %IX10.3 : BOOL; // motor is on
PIm_error AT %IX10.4 : BOOL; // error state; reset with P0reset
PIs_v AT %IW6 : INT; // v [v]=1cm/s

```

Modbus Coils map to BOOL and Modbus Registers map to 16bit word for other values. Modbus/TCP can be implemented in Java by means of <https://sourceforge.net/projects/easymodbustcp-udp-java/>. A version with pom.xml for Maven builds can be found at [2].

4.2 Distances and sensor positions

This table describes sensor positions relative to each level. If a range is given, it describes the tolerance of detection.

Description/Name	Value	Range (+/-)
Level height	3,5m	/
*al	-1,5m	50mm
*sl	-50mm	5mm
*r	0mm	5mm
*su	50mm	5mm
*au	1.5m	50mm

4.3 Speeds

$$|v_1| \approx 0,1 \frac{m}{s}$$

$$|v_2| \approx 1 \frac{m}{s}$$

5 More information / links

- [1] Documentation of HW and SW of the laboratory of automation technology at the HTWG Konstanz
- [2] Modbus-TCP Library adapted for Maven https://gitlab.ei.htwg-konstanz.de/system-architecture/25ws/easymodbus-maven/-/tree/ei-main?ref_type=heads